3. Distributions

Ω domain in $\mathbb{R}^n$. (bounded or not)

$\mathcal{D}(\Omega) = C_0^\infty(\Omega) = \{ u \in C^\infty(\Omega) \mid \overline{\operatorname{supp} u} \text{ is compact in } \Omega \}$

$\operatorname{supp} u = \{ x \in \Omega \mid u(x) \neq 0 \}$ $\quad \Leftrightarrow \operatorname{supp} u$

$\Leftrightarrow \exists$ cpt K, $\operatorname{supp} u \subset K \subset \Omega$

If $u \in \mathcal{D}(\Omega)$, u is compactly supported.

Ex 1 The standard molifier

$$j(x) = \begin{cases} C e^{\frac{1}{|x|^2-1}} & x \in \mathbb{R}^n,\ |x| < 1 \\ 0 & x \in \mathbb{R}^n,\ |x| \geq 1 \end{cases}$$

where C is chosen s.t. $\int_{\mathbb{R}^n} j(x)\,dx \geq 1$.

Claim. $j(x) \in \mathcal{D}(\mathbb{R}^n) = C_0^\infty(\mathbb{R}^n)$

Pf. $\operatorname{supp} j(x) = B_1(0)$ (open) $\Rightarrow j(x)$ is compactly supported

Consider $f(t) = \begin{cases} C e^{\frac{1}{t-1}} & 0 < t < 1 \\ 0 & t \geq 1 \end{cases} \Rightarrow j(x) = f(|x|^2)$

To show $j(x) \in C^\infty(\mathbb{R}^n)$, just need to show $f \in C^\infty([0,\infty))$

How about $t = 1$? $f'(1^+) = 0$ $\quad f'(1^-) = \lim_{t \to 1^-} \frac{f(t) - f(1)}{t-1}$

$$= \lim_{t \to 1} \frac{C e^{\frac{1}{t-1}}}{t-1}$$

$\Rightarrow f'(1)$ exists and $f'(1) = 0$ $\quad = \lim_{s \to -\infty} C e^s \cdot s$

$f''(1^+) = 0$

$$f''(1^-) = \lim_{t \to 1^-} \frac{f'(t) - f'(1)}{t-1} = \lim_{t \to 1^-} \frac{C e^{\frac{1}{t-1}} (-1)(t-1)^{-2}}{t-1} = \lim_{t \to 1^-} \frac{-C e^{\frac{1}{t-1}}}{(t-1)^3}$$

$$= \lim_{s \to -\infty} -C e^s \cdot s^3 = 0$$

Similarly $f^{(k)}(1) = 0$ $\forall k \geq 1 \Rightarrow f \in C^\infty([0,\infty))$ $\square$

$(C^\infty(\Omega) \neq C^\omega(\Omega))$ The molifier is not analytic in Ω

Is $\mathcal{D}(\Omega) \neq \emptyset$?

Fix a point $a \in \Omega$. Take $\varepsilon > 0$ small such that $\overline{B_\varepsilon(a)} \subset \Omega$

Let $j_{\varepsilon,a}(x) = \frac{1}{\varepsilon^n} j\left(\frac{x-a}{\varepsilon}\right)$ $\quad \left|\frac{x-a}{\varepsilon}\right| < 1 \iff |x-a| < \varepsilon$

$\Rightarrow j_{\varepsilon,a} \in \mathcal{D}(\Omega)$. $\operatorname{supp} j_{\varepsilon,a} = B_\varepsilon(a)$ (open)

$$\& \quad \int_\Omega j_{\varepsilon,a}(x)\,dx \underset{dy = (\frac{1}{\varepsilon})^n dx}{\overset{y = \frac{x-a}{\varepsilon}}{=\!=}} \int_{\Omega \to \mathbb{R}^n} \frac{1}{\varepsilon^n} j\left(\frac{x-a}{\varepsilon}\right) dx$$

$$= \int_{\mathbb{R}^n} \frac{1}{\varepsilon^n} j(y)\, \varepsilon^n\, dy = 1 \qquad \square$$

<u>Definition.</u> A distribution on $\mathcal{D}(\Omega)$ is a linear functional

$$f : \mathcal{D}(\Omega) \longrightarrow \mathbb{R}. \quad \varphi \longmapsto f(\varphi) = \langle f, \varphi \rangle \text{ Notation}$$

s.t. f is continuous on $\mathcal{D}(\Omega)$ in the following sense

For any $\{\varphi_k\} \subset \mathcal{D}(\Omega)$, s.t

(i) $\operatorname{supp} \varphi_k \subset$ fixed compact $K \in \Omega$

(ii) $\forall \alpha \in (\alpha_1, \cdots, \alpha_n) \in \mathbb{Z}^n$, $\alpha_i \geq 0$

$$\lim_{k \to \infty} \|\partial^\alpha \varphi_k\|_{L^\infty(\Omega)} = 0 \quad \text{where } \partial^\alpha = \partial_{x_1}^{\alpha_1} \partial_{x_2}^{\alpha_2} \cdots \partial_{x_n}^{\alpha_n}$$

we have $\langle f, \varphi_k \rangle \to 0$ as $k \to \infty$.

f is continuous at $\varphi \equiv 0$. $\forall \varphi_k \to 0$ in $\mathcal{D}(\Omega)$

WTS $\underset{\substack{\| \\ f(\varphi_k)}}{\langle f, \varphi_k \rangle} \to \underset{\substack{\| \\ f(0) = 0}}{\langle f, 0 \rangle}$ $\qquad \mathcal{D}(\Omega) = C_0^\infty(\Omega)$

$\mathcal{D}(\Omega) = X$. $\mathcal{D}'(\Omega)$ = set of all distributions on $\mathcal{D}(\Omega)$,

$= X'$ (dual space)

Ex. 2 Let $f \in L^1_{loc}(\Omega)$ (i.e. $\forall$ compact $K \subset \Omega$. $f \in L^1(K)$, i.e $\int_K |f| dx < \infty$)

Define $F \in \mathcal{D}(\Omega)$ as $\langle F, \varphi \rangle \triangleq \int_\Omega f \varphi \, dx \in \mathbb{R}$

Claim. $F \in \mathcal{D}'(\Omega)$. Suppose $\{\varphi_k\}_{k=1}^{\infty}$ satisfies conditions (i) and (ii) in the above definition

$$|\langle F, \varphi_k \rangle| = \left| \int_\Omega f \varphi_k \, dx \right| \leq \int_K |f(x)| \, \|\varphi_k\|_{L^\infty(\Omega)} dx \xrightarrow{k \to \infty} 0.$$

(by (ii) with $\alpha = (0, \cdots, 0)$ $\Rightarrow \|\varphi_k\|_{L^\infty(\Omega)} \to 0$)

F is called the distribution induced by $f(x)$ (Often write $F = f$)

Ex. 3. Suppose $f \in \mathcal{D}'(\Omega)$, $g \in C^\infty(\Omega)$

Define: $\langle f g, \varphi \rangle = \langle f, \underbrace{g \varphi}_{\in C_0^\infty(\Omega) = \mathcal{D}(\Omega)} \rangle \quad \forall \varphi \in \mathcal{D}(\Omega)$

$\Rightarrow fg := gf \in \mathcal{D}'(\Omega)$ (distr. C^∞ = distr)

Ex. 4. Fix an $a \in \Omega$. define

$\delta_a \; \mathcal{D}(\Omega) \to \mathbb{R}$, $\langle \delta_a, \varphi \rangle = \varphi(a) \Rightarrow \delta_a \in \mathcal{D}'(\Omega)$

δ-function $\delta_a(x) = \begin{cases} \infty & x = a \\ 0 & x \neq a \end{cases}$ and $\int_{\mathbb{R}^n} \delta_a(x) dx = 1$

$\Rightarrow \int_{\mathbb{R}^n} \delta_a(x) \varphi(x) dx = \varphi(a)$ if φ is continuous at $x = a$

- δ_a is the Dirac measure giving the unit mass to the point a
- Discrete case Kronecker delta $\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$

<u>Definition</u> Let $\{f_k\}_{k=1}^{\infty} \subset \mathcal{D}'(\Omega)$, $f \in \mathcal{D}'(\Omega)$. we say $f_k \to f$ as $k \to \infty$ if $\forall \varphi \in \mathcal{D}(\Omega)$, $\langle f_k, \varphi \rangle \to \langle f, \varphi \rangle$ as $k \to \infty$.

<u>Theorem</u> (Spiky) Let $\{f_k\}_{k=1}^{\infty} \subset L^1(\Omega)$, s.t

(i) $\{f_k\}$ concentrates at $a \in \Omega$: $\forall$ small $\delta > 0$

$$\int_{\Omega \setminus B_\delta(a)} |f_k| \, dx \xrightarrow{k \to \infty} 0$$

(ii) $\|f_k\|_{L^1(\Omega)} \leq$ const. M. $\forall k \geq 1$.

(iii) $\int_\Omega f_k(x) \, dx \to A$ as $k \to \infty$

Then f_k (understood as distributions) $\longrightarrow A \delta_a$ as $k \to \infty$

Proof: Just need to show $\forall \varphi \in \mathcal{D}(\Omega)$

$$\langle f_k(x), \varphi \rangle \xrightarrow{k \to \infty} \langle A\delta_a, \varphi \rangle$$

$$\| \qquad\qquad\qquad \|$$

$$\int_\Omega f_k \varphi \, dx \qquad\qquad A\varphi(a)$$

$$\text{LHS} = \int_\Omega f_k(x) (\varphi_k(x) - \varphi(a)) \, dx + \int_\Omega f_k(x) \varphi(a) \, dx$$

Since φ is continuous at $x = a$, $\forall \varepsilon > 0 \; \exists \delta > 0$ s.t

$$|\varphi(a) - \varphi(x)| < \varepsilon \quad \text{if } |x - a| < \delta, \; x \in \Omega.$$

Now $|I_k| \leq \int_{\Omega \setminus B_\delta(a)} | \quad | \, dx + \int_{\Omega \cap B_\delta(a)} | \quad | \, dx$

$$\leq 2 \|\varphi\|_{L^\infty(\Omega)} \cdot \int_{\Omega \setminus B_\delta(a)} |f_k(x)| \, dx$$

$$+ \underbrace{\varepsilon \int_{\Omega \cap B_\delta(a)} |f_k| \, dx}_{\leq \varepsilon M}$$

Take $\limsup_{k\to\infty}$ of both sides

$\overset{(1)}{\Longrightarrow} \limsup_{k\to\infty} |I_k| \le 0 + M\varepsilon$. ε arbitrary

$\Longrightarrow \lim_{k\to\infty} |I_k| = 0$ $\square$

Ex. 5 $\triangle$ Fix an $a \in \Omega$, then $j_{\varepsilon,a} \to \delta_a$ as $\varepsilon \to 0$ in the sense of distributions. $(k = \frac{1}{\varepsilon})$

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Review $\Omega \subset \mathbb{R}^n$ bounded or not

$\mathcal{D}(\Omega) = C_0^\infty(\Omega)$, $\mathcal{D}'(\Omega) = \{ f : \mathcal{D}(\Omega) \to \mathbb{R} \mid f$ is linear and continuous$\}$

f is continuous at $\varphi = 0$.

$$(*) \begin{cases} \forall \{\varphi_k\} \subset \mathcal{D}(\Omega) \quad \operatorname{supp} \varphi_k \subset K \text{ compact}, \forall k \\ \|D^\alpha \varphi_k\|_{L^\infty(\Omega)} \xrightarrow{k\to\infty} 0 \quad \forall \alpha = (\alpha_1, \dots, \alpha_n),\ \alpha_i \in \mathbb{Z}, \alpha_i \ge 0 \\ \Rightarrow f(\varphi_k) (= \langle f, \varphi_k \rangle) \to 0. \end{cases}$$

Derivatives of distributions

Motivation $f \in C^1(\Omega) \Rightarrow f, f_{x_i} \in L^1_{loc}(\Omega)$

$\left(\forall K \text{ compact in } \Omega \right.$
$\left. \int_K |f|\, dx < \infty,\ \int_K |f_{x_i}|\, dx < \infty \right)$.

$$\langle f_{x_i}, \varphi \rangle = \int_\Omega f_{x_i} \varphi\, dx = \int_\Omega [(f\varphi)_{x_i} - f\varphi_{x_i}]\, dx$$

$$= \underbrace{\cancel{\int_{\partial\Omega} f\varphi \nu_i\, d\sigma}}_{\text{since } \varphi \in C_0^\infty(\Omega)} - \int_\Omega f\varphi_{x_i}\, dx$$

$$= -\int_\Omega f \varphi_{x_i}\, dx$$

<u>Definition</u> $\forall f \in \mathcal{D}'(\Omega)$, define its distributional derivatives

$$\frac{\partial f}{\partial x_i} (= f_{x_i} = D_{x_i} f = D_i f) \text{ by}$$

$$\langle D_i f, \varphi \rangle \triangleq -\langle f, \varphi_{x_i} \rangle, \ \forall \varphi \in \mathcal{D}(\Omega)$$

<u>Fact</u> $D_i f \in \mathcal{D}'(\Omega)$

Linearity: $\langle D_i f, \alpha_1 \varphi_1 + \alpha_2 \varphi_2 \rangle = -\langle f, (\alpha_1 \varphi_1 + \alpha_2 \varphi_2)_{x_i} \rangle$

$$= -\langle f, \alpha_1 (\varphi_1)_{x_i} + \alpha_2 (\varphi_2)_{x_i} \rangle$$

$$= -\left[\alpha_1 \langle f, (\varphi_1)_{x_i} \rangle + \alpha_2 \langle f, (\varphi_2)_{x_i} \rangle\right]$$

$$= \alpha_1 \langle D_i f, \varphi_1 \rangle + \alpha_2 \langle D_i f, \varphi_2 \rangle$$

$D_i f$ is continuous at $\varphi = 0$: Take $\{\varphi_k\} \subset \mathcal{D}(\Omega)$ as in $(*)$,

$$\langle D_i f, \varphi_k \rangle = -\langle f, (\varphi_k)_{x_i} \rangle = -\langle f, \tilde{\varphi}_k \rangle \xrightarrow{k \to \infty} 0$$

(since $\{\tilde{\varphi}_k\}$ satisfies $*$, and f is continuous at $\varphi = 0$)

In general, $\forall \alpha \in \mathbb{Z}^n$, $\alpha = (\alpha_1, \dots, \alpha_n)$, $\alpha_i \geq 0$ define the distributional derivative $D^\alpha f$ by

$$\langle D^\alpha f, \varphi \rangle = (-1)^{|\alpha|} \langle f, D^\alpha \varphi \rangle, \ \forall \varphi \in \mathcal{D}(\Omega), \ (|\alpha| = \alpha_1 + \dots + \alpha_n)$$

$\Rightarrow D^\alpha f \in \mathcal{D}'(\Omega)$

Ex. Heaviside function $H(x) = \begin{cases} 0, & x < 0 \\ 1 & x \geq 0 \end{cases}$

- $H(x) \in C^1(\mathbb{R})$
- $H \in L^1_{loc}(\mathbb{R})$

Question:

$$DH(x) = H'(x) \text{ (in the sense of distribution)} = ?$$

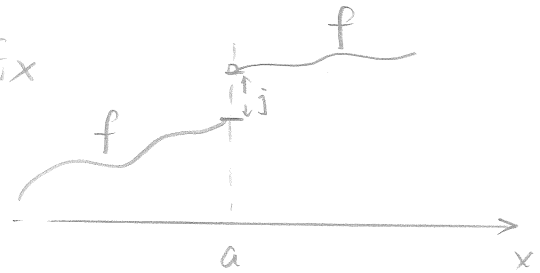
$\forall \varphi \in \mathcal{D}(\mathbb{R})$

$$\langle DH, \varphi \rangle = -\langle H, \varphi'(x) \rangle = -\int_{\mathbb{R}} H(x)\varphi'(x)\,dx$$

$$= \int_0^{\infty} \varphi'(x)\,dx = -\varphi(x)\Big]_0^{\infty} = \varphi(0)$$

$$\Longrightarrow DH = \delta_0$$

Ex



$f \in L^1_{loc}(\mathbb{R})$

$f \in C^1((-\infty, a]) \cup C^1((a, +\infty))$

Question:

$Df = ?$ (in the sense of distribution)

$$Df = \underbrace{f'(x)}_{\in L^1_{loc}(\mathbb{R})} + j\,\delta_a(x) \qquad f(x) \begin{cases} f'(x) \text{ classical derivatives} \\ \text{any value } x = a \end{cases}$$

$\forall \varphi \in \mathcal{D}(\mathbb{R})$

$$\langle Df, \varphi \rangle \overset{def}{=} -\langle f, \varphi'(x) \rangle = -\int_{\mathbb{R}} f\varphi'(x)\,dx$$

$$= -\int_{-\infty}^{a} f\varphi'\,dx - \int_a^{\infty} f\varphi'\,dx$$

$$= (f\varphi)(-\infty) - (f\varphi)(a^-) + \int_{-\infty}^{a} f'\varphi\,dx$$

$$+ (f\varphi)(a^+) - (f\varphi)(\infty) + \int_a^{\infty} f'\varphi\,dx$$

$$= \int_{\mathbb{R}} f'\varphi\,dx + [f(a^+) - f(a^-)]\,\varphi(a)$$

$$= \langle f', \varphi \rangle + \langle j\,\delta_a, \varphi \rangle$$

Distributional solutions of PDEs

Consider $L : \mathcal{D}'(\Omega) \to \mathcal{D}'(\Omega)$

$$u \mapsto Lu \triangleq \sum_{|\alpha|=0}^{m} A_\alpha(x) D^\alpha u \,.\; A_\alpha(x) \in C^\infty(\Omega)$$

(in the sense of distribution)

$(f \in \mathcal{D}'(\Omega) \;.\; g \in C^\infty(\Omega)$

e.g.) $fg = gf \in \mathcal{D}'(\Omega)$

$\langle fg, \varphi \rangle \triangleq \langle f, g\varphi \rangle \,.\; \forall \varphi \in C_0^\infty(\Omega)$

$\forall \varphi \in \mathcal{D}(\Omega)$

$$\langle Lu, \varphi \rangle = \langle \sum_{|\alpha|=0}^{m} A_\alpha(x) D^\alpha u, \varphi \rangle$$

$$= \sum_{|\alpha|=0}^{m} \langle A_\alpha(x) D^\alpha u, \varphi \rangle$$

$$= \sum_{|\alpha|=0}^{m} \langle D^\alpha u, A_\alpha(x) \varphi \rangle$$

$$= \sum_{|\alpha|=0}^{m} (-1)^\alpha \langle u, D^\alpha [A_\alpha(x) \varphi] \rangle$$

$$= \langle u, \underbrace{\sum_{|\alpha|=0}^{m} (-1)^\alpha D^\alpha [A_\alpha(x) \varphi]}_{\triangleq L^* \varphi} \rangle$$

$$\langle Lu, \varphi \rangle = \langle u, L^* \varphi \rangle$$

L^* adjoint operator of L

$L^* \quad \mathcal{D}(\Omega) \to \mathcal{D}(\Omega)$

Ex. $L = \Delta$ Question: $L^* = ?$

$m = 2 \quad A_\alpha(x) = \begin{cases} 1 & \alpha_i = 2 \text{ for some } i = \{1, 2, \dots, n\}, \; \alpha_j = 0 \text{ if } j \neq i \\ 0 & \text{otherwise} \end{cases}$

$$L^* \varphi = \sum_{i=1}^{n} (-1)^2 D_{x_i x_j} \varphi = \Delta \varphi$$

$L = \Delta : \mathcal{D}'(\Omega) \to \mathcal{D}'(\Omega)$ (in the sense of distribution)

$L^* = \Delta : \mathcal{D}(\Omega) \to \mathcal{D}(\Omega)$ (classical derivative)

We say Δ is symmetric. : $\langle \Delta u, \varphi \rangle = \langle u, \Delta \varphi \rangle$

<u>Definition</u> Consider PDE

$$Lu = f \quad f \in \mathcal{D}'(\Omega) \text{ given.}$$

(e.g. $-\Delta u = f(x)$)

If $u \in \mathcal{D}'(\Omega)$ and $Lu = f$ in the sense of distribution,

i.e. $\forall \varphi \in \mathcal{D}(\Omega)$

$$\langle Lu, \varphi \rangle = \langle f, \varphi \rangle$$

$$\| $$

$$(\langle u, L^* \varphi \rangle)$$

then we say u is a distributional solution of the PDE.

Remark: Classical solution must be a distributional solution.

Ex. $-\Delta u = \delta_y$ in $\mathbb{R}^n$, $y \in \mathbb{R}^n$, fixed $(*1)$

<u>Definition</u> A distributional solution of $(*1)$ is called a fundamental solution (F.S)

Question Uniqueness? (No. If u is a FS, then $u + c$ is also an FS)

Recall: $$u(x) = \begin{cases} -\frac{1}{2\pi} \ln|x-y| & n \geq 2 \quad (C_1 \ln r) \\ \frac{1}{(n-2)\omega_n |x-y|^{n-2}} & n \geq 3 \quad (C_2 \frac{1}{r^{n-2}}) \end{cases}$$

ω_n: area of unit sphere in $\mathbb{R}^n$

Satisfies $-\Delta u = 0$ in $\mathbb{R}^n \setminus \{y\}$